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United States Patent Application Publication

20250280535

Kind Code

A1

Publication Date

September 04, 2025

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SINGLE-POLY NON-VOLATILE MEMORY CELL

Abstract

A single-poly non-volatile memory cell is provided. In the memory cell, a polysilicon gate layer of a gate structure is used as a floating gate of the floating gate transistor. During an ion implantation process, a relatively small amount of dopant is implanted into the polysilicon gate layer. Consequently, the dopant concentration of the polysilicon gate layer is less than the dopant concentration of a merged n-type doped region beside the gate structure. Since the defects in the polysilicon gate layer are reduced, the data retention capability of the memory cell is effectively enhanced.

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Family ID: 96828116

Appl. No.: 19/006303

Filed: December 31, 2024

Related U.S. Application Data

us-provisional-application US 63559234 20240229

us-provisional-application US 63682408 20240813

Publication Classification

Int. Cl.: H10B41/60 (20230101); H10B41/35 (20230101)

U.S. Cl.:

CPC H10B41/60 (20230201); H10B41/35 (20230201);

Background/Summary

[0001] This application claims the benefit of U.S. provisional application Ser. No. 63/559,234, filed Feb. 29, 2024, and the benefit of U.S. provisional application Ser. No. 63/682,408, filed Aug. 13, 2024, the subject matters of which are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to a memory cell of a memory, and more particularly to a single-poly non-volatile memory cell.

BACKGROUND OF THE INVENTION

[0003] As is well known, a non-volatile memory includes an array structure. The array structure is composed of a plurality of non-volatile memory cells. Each non-volatile memory cell includes a storage unit. For example, the storage unit is a floating gate transistor. The storage state of the non-volatile memory cell is determined according to the number of charges stored in the floating gate of the floating gate transistor.

[0004] In order to be compatible with the standard CMOS manufacturing process, the memory cell of the conventional non-volatile memory is equipped with a single-poly floating gate transistor. The floating gate transistor and associated electronic devices are collaboratively formed as a single-poly non-volatile memory cell.

[0005] For example, an erasable programmable single-poly non-volatile memory is disclosed in U.S. Pat. No.8,941,167.

SUMMARY OF THE INVENTION

[0006] An embodiment of the present invention provides a single-poly non-volatile memory cell. The single-poly non-volatile memory cell includes an isolation structure, a first well region, a second well region, a first gate structure, a second gate structure, a first spacer, a second spacer, a first merged doped region, a second merged doped region, a third merged doped region and a fourth merged doped region. The isolation structure is formed on a substrate. The first well region is formed on the substrate. The second well region is formed on the substrate. The isolation structure is located between the first well region and the second well region. The first gate structure and a second gate structure are formed on a surface of the first well region. The second gate structure is extended to the second well region through a surface of the isolation structure. A portion of the second well region is covered by the second gate structure. The first spacer is formed on a sidewall of the first gate structure. The second spacer is formed on a sidewall of the second gate structure. The first merged doped region is formed under a surface of the first well region and located beside a first side of the first gate structure. The second merged doped region is formed under the surface of the first well region. The second merged doped region is arranged between a second side of the first gate structure and a first side of the second gate structure. The third merged doped region is formed under the surface of the first well region and located beside a second side of the second gate structure. The fourth merged doped region is formed under a surface of the second well region and located beside the second gate structure. The first gate structure includes a first gate oxide layer and a first polysilicon gate layer. The second gate structure includes a second gate oxide layer and a second polysilicon gate layer. A dopant concentration of the second polysilicon gate layer above the second well region is less than a dopant concentration of the fourth merged doped region.

[0007] Numerous objects, features and advantages of the present invention will be readily apparent upon a reading of the following detailed description of embodiments of the present invention when taken in conjunction with the accompanying drawings. However, the drawings employed herein are for the purpose of descriptions and should not be regarded as limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The above objects and advantages of the present invention will become more readily apparent to those ordinarily skilled in the art after reviewing the following detailed description and accompanying drawings, in which:

[0009] FIGS. 1A to 1K schematically illustrate the steps of a method of manufacturing a single-poly non-volatile memory cell according to a first embodiment of the present invention;

[0010] FIG. 1L is a schematic equivalent circuit diagram of the single-poly non-volatile memory cell according to the first embodiment of the present invention;

[0011] FIG. 2A is a schematic circuit diagram the bias voltages of performing a program action on the memory cell according to the first embodiment of the present invention;

[0012] FIG. 2B is a schematic circuit diagram the bias voltages of performing an erase action on the memory cell according to the first embodiment of the present invention;

[0013] FIG. 2C is a schematic circuit diagram the bias voltages of performing a read action on the memory cell according to the first embodiment of the present invention;

[0014] FIGS. 3A to 3C schematically illustrate the steps of a method of manufacturing a single-poly non-volatile memory cell according to a second embodiment of the present invention; and

[0015] FIGS. 4A to 4D schematically illustrate the steps of a method of manufacturing a single-poly non-volatile memory cell according to a third embodiment of the present invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0016] FIGS. 1A to 1K schematically illustrate the steps of a method of manufacturing a single-poly non-volatile memory cell according to a first embodiment of the present invention. FIG. 1L is a schematic equivalent circuit diagram of the single-poly non-volatile memory cell according to the first embodiment of the present invention. Hereinafter, the single-poly non-volatile memory cell is referred as a memory cell.

[0017] As shown in FIG. 1A, an isolation structure forming step is performed. Firstly, an isolation structure such as a shallow trench isolation (STI) structure **102** is formed on a substrate sub. Due to the isolation structure **102**, a region A and a region B is defined. For example, the substrate sub is a p-type substrate, an n-type substrate, a glass substrate, or any other appropriate substrate. That is, the substrate sub is covered by the isolation structure **102**, but only the surface of the p-type substrate corresponding to the region A and the region B is exposed. In the memory cell of the first embodiment, two series-connected p-type transistors are formed in the region A, and an n-type transistor is formed in the region B.

[0018] Then, as shown in FIG. 1B, a first well region forming step and a second well region forming step are performed. In the first well region forming step, the region A is exposed, and the region B is covered. Consequently, a first well region is formed. For example, the first well region is an N-well region. In other words, the N-well region is formed under the surface of the region A. In the second well region forming step, the region B is exposed, and the region A is covered. Consequently, a second well region is formed. For example, the second well region is a P-well region. In other words, the P-well region is formed under the surface of the region B. It is noted that the sequence of the first well region forming step and the second well region forming step is not restricted.

[0019] FIG. 1C is a schematic cross-sectional view illustrating the resulting structure of FIG. 1B and taken along with the line a-b. As shown in FIG. 1C, the N-well region is formed under the surface of the region A, the P-well region is formed under the surface of the region B, and the isolation structure **102** is located between the N-well region and the P-well region. In addition, the portion of the N-well region and the portion of the P-well region underlying the isolation structure **102** are in contact with each other.

[0020] As shown in FIG. 1D, a gate structure forming step is performed. Firstly, two gate oxide layers **103** and **105** are formed. Then, two polysilicon gate layers **113** and **115** are formed on the two gate oxide layers **103** and **105**, respectively. Consequently, two gate structures **123** and **125** are formed.

[0021] Please refer to FIG. 1D again. The two gate structures **123** and **125** are formed on the surface of the region A. In addition, the region A is divided into three sub-regions by the two gate structures **123** and **125**. The gate structure **125** is extended externally from the region A to the region B through the surface of the isolation structure **102**. In addition, a portion of the region B is covered by the gate structure **125**. The polysilicon gate layer **115** of the gate structure **125** is served as a floating gate. The polysilicon gate layer **113** of the gate structure **123** is served as a select gate.

[0022] Subsequently, a doped region forming step and a conducting line forming step are performed. In the following drawings, the cross-sectional view of FIG. 1D and taken along the line c-d will be shown to describe the doped region forming step and the conducting line forming step.

[0023] As shown in FIG. 1E, a photoresist layer **130** indicated by dashed lines is formed to cover the region B and the overlying gate structure **125**. Then, a p-type lightly doped drain process (PLDD process) is performed on the region A with the gate structures **123** and **125** serving as the hard masks. Consequently, three p-type lightly doped drain regions (PLDD regions) **141**, **142** and **143** are formed under the surface of the N-well region NW corresponding to the region A. The PLDD region **141** is formed under the surface of the N-well region NW and located beside the first side of the gate structure **123**. The PLDD region **142** is formed under the surface of the N-well region NW and arranged between the second side of the gate structure **123** and the first side of the gate structure **125**. The PLDD region **143** is formed under the surface of the N-well region NW and located beside the second side of the gate structure **125**.

[0024] Please refer to FIG. 1F. After the photoresist layer **130** is removed, a photoresist layer **132** indicated by dashed lines is formed to cover the region A and the overlying gate structures **123** and **125**. Then, an n-type lightly doped drain process (NLDD process) is performed on the region B with the gate structure **125** serving as the hard mask. Consequently, an n-type lightly doped drain region (NLDD region) **145** is formed under the surface of the P-well region PW corresponding to the region B. In addition, the NLDD region **145** is located beside the gate structure **125**.

[0025] Please refer to FIG. 1G. After the photoresist layer **132** is removed, a spacer **153** is formed on the sidewall of the gate structure **123**, and a spacer **155** is formed on the sidewall of the gate structure **125**.

[0026] As shown in FIG. 1H, a photoresist layer **134** indicated by dashed lines is formed to cover the region B and the overlying gate structure **125** and spacer **155**. Please refer to FIG. 1H. Then, a p-type ion implantation process is performed on the region A by using the two gate structures **123** and **125** and the two spacers **153** and **155** as hard masks. Consequently, three p-type ion implantation regions **161**, **162** and **163** indicated by oblique lines are formed on three sub-regions of the region A uncovered by the two gate structures **123** and **125** and the two spacers **153** and **155**. Especially, the dopant concentration of the p-type ion implantation regions **161**, **162** and **163** is greater than the dopant concentration of the PLDD regions **141**, **142** and **143**. The p-type ion implantation regions **161**, **162** and **163** are the source/drain region.

[0027] Please refer to FIG. 1H again. The PLDD region **141** and the p-type ion implantation region **161** are partially overlapped with each other. In addition, the PLDD region **141** and the p-type ion implantation region **161** are collaboratively formed as a merged p-doped region **171**. The PLDD region **142** and the p-type ion implantation region **162** are partially overlapped with each other. In addition, the PLDD region **142** and the p-type ion implantation region **162** are collaboratively formed as a merged p-doped region **172**. The PLDD region **143** and the p-type ion implantation region **163** are partially overlapped with each other. In addition, the PLDD region **143** and the p-type ion implantation region **163** are collaboratively formed as a merged p-doped region **173**. The merged p-doped region **171** is formed under the surface of the N-well region NW and located

beside the first side of the gate structure **123**. The merged p-doped region **172** is formed under the surface of the N-well region NW and arranged between the second side of the gate structure **123** and the first side of the gate structure **125**. The merged p-doped region **173** is formed under the surface of the N-well region NW and located beside the second side of the gate structure **125**.

[0028] Furthermore, when the p-type ion implantation process is performed, the two gate structures **123** and **125** are used as the hard masks. Consequently, the polysilicon gate layers **113** and **115** over the region A are formed as the p-type polysilicon gate layers. The dopant concentration of the p-type polysilicon gate layers is equal to the dopant concentration of the merged p-doped region **173**.

[0029] Please refer to FIG. **11**. After the photoresist layer **134** is removed, a photoresist layer **136** indicated by dashed lines is formed to cover the region A, the overlying gate structures **123**, **125** and the overlying spacers **153**, **155**. Then, an n-type ion implantation process is performed on the region B by using the gate structure **125** and the spacer **155** as hard masks.

[0030] Consequently, an n-type ion implantation region **165** indicated by oblique lines is formed in the region B uncovered by the gate structure **125** and the spacer **155**. Especially, the dopant concentration of the n-type ion implantation region **165** is greater than the dopant concentration of the NLDD region **145**.

[0031] Please refer to FIG. **11** again. The NLDD region **145** and the n-type ion implantation region **165** are partially overlapped with each other. In addition, the NLDD region **145** and the n-type ion implantation region **165** are collaboratively formed as a merged n-doped region **175**. In addition, the merged n-doped region **175** is located beside the gate structure **125**.

[0032] Furthermore, when the n-type ion implantation process is performed, the gate structure **125** is used as the hard mask. Consequently, the polysilicon gate layer **115** over the region B is formed as the n-type polysilicon gate layers. The dopant concentration of the n-type polysilicon gate layer is equal to the dopant concentration of the merged n-doped region **175**.

[0033] Please refer to FIG. **1J**. After the photoresist layer **136** is removed and the conducting line forming step is completed, the memory cell of the first embodiment is fabricated.

[0034] That is, the merged p-type doped region **171** is connected with a source line SL, the merged p-type doped region **173** is connected with a bit line BL, the merged n-type doped region **175** is connected with an erase line EL, and the polysilicon gate layer **113** is connected with a select gate line SG. The three-dimensional view of the resulting structure of FIG. **1J** is shown in FIG. **1K**.

[0035] In the region A, the gate structure **123** and the merged p-type doped regions **171** and **172** on its two sides are collaboratively formed as a select transistor. In addition, the gate structure **125** and the two merged p-doped regions **172** and **173** on its two sides are collaboratively formed as a floating gate transistor. The polysilicon gate layer **115** of the gate structure **125** is the floating gate of the floating gate transistor. Furthermore, the floating gate transistor and the select transistor are p-type transistors.

[0036] As mentioned above, the gate structure **125** is extended externally from the region A to the surface of the region B. Consequently, the gate structure **125** and the merged n-doped region **175** corresponding to the region B are collaboratively formed as an n-type transistor. The two drain/source terminals of the n-type transistor are connected with each other. Consequently, the n-type transistor is connected as an n-type MOS capacitor.

[0037] As shown in FIG. **1L**, the memory cell includes a select transistor M.sub.S, a floating gate transistor M.sub.F and a MOS capacitor C.sub.MOS. The gate terminal of the select transistor M.sub.S is connected with the select gate line SG. The first drain/source terminal of the select transistor M.sub.S is connected with the source line SL. The first drain/source terminal of the floating gate transistor M.sub.F is connected with the second drain/source terminal of the select transistor M.sub.S. The second drain/source terminal of the floating gate transistor M.sub.F is connected with the bit line BL. The first terminal of the MOS capacitor CMOS is connected with the floating gate **115** of the floating gate transistor M.sub.F. The second terminal of the MOS capacitor C.sub.MOS is connected with the erase line EL.

[0038] FIG. 2A is a schematic circuit diagram the bias voltages of performing a program action on the memory cell according to the first embodiment of the present invention. FIG. 2B is a schematic circuit diagram the bias voltages of performing an erase action on the memory cell according to the first embodiment of the present invention. FIG. 2C is a schematic circuit diagram the bias voltages of performing a read action on the memory cell according to the first embodiment of the present invention.

[0039] Please refer to FIG. 2A. When the program action is performed, the source line SL receives a program voltage $V_{sub.PP}$, the select gate line SG receives an on voltage $V_{sub.ON}$, the bit line BL receives a ground voltage (0V), and the erase line EL receives the ground voltage (0V). For example, the program voltage $V_{sub.PP}$ is 9V, and the on voltage $V_{sub.ON}$ is 0V.

[0040] When the program action is performed, the select transistor $M_{sub.S}$ is turned on, and a program current $I_{sub.P}$ is generated between the bit line BL and the source line SL. Consequently, when the hot carriers (e.g., electrons) of the program current $I_{sub.P}$ flow through a channel region corresponding to the floating gate **115**, a channel hot electron effect (CHE effect) is generated. Due to the CHE effect, electrons are injected into the floating gate **115**.

[0041] Please refer to FIG. 2B. When the erase action is performed, the source line SL receives the ground voltage (0V), the bit line BL receives the ground voltage (0V), the select gate line SG receives the ground voltage (0V), and the erase line EL receives an erase voltage $V_{sub.EE}$. For example, the erase voltage $V_{sub.EE}$ is 12V.

[0042] When the erase action is performed, the select transistor $M_{sub.S}$ is turned off. Under this circumstance, a Fowler-Nordheim Tunneling (FN) effect is generated at the two terminals of the MOS transistor $C_{sub.MOS}$. Consequently, hot carriers are ejected from the floating gate **115** to the erase line EL.

[0043] Please refer to FIG. 2C. When the read action is performed, the source line SL receives a read voltage $V_{sub.R}$, the bit line BL receives the ground voltage (0V), the select gate line SG receives the on voltage $V_{sub.ON}$, and the erase line EL receives the ground voltage (0V). For example, the read voltage $V_{sub.R}$ is 5V.

[0044] When a read action is performed, the select transistor $M_{sub.S}$ is turned on, and a read current $I_{sub.R}$ is generated between the bit line BL and the source line SL. The storage state of the memory cell can be determined according to the magnitude of the read current $I_{sub.R}$. For example, in case that no electrons are stored in the floating gate **115**, the magnitude of the read current I_R is very low (e.g., nearly zero). Consequently, it is determined that the memory cell is in a first storage state. Whereas, in case that electrons are stored in the floating gate **115**, the magnitude of the read current $I_{sub.R}$ is very high. Under this circumstance, it is determined that the memory cell is in a second storage state.

[0045] In the ion implantation process of manufacturing the memory cell of

[0046] the first embodiment, ions are accelerated to the high energy state. Consequently, ions are impinged on a target and implanted in the target.

[0047] Generally, when the n-type ion implantation process is performed, arsenic ions are used as dopant. For example, when the n-type ion implantation process is performed on the resulting structure of FIG. 1I, the gate structure **125** is used as the hard mask, and arsenic ions are implanted into the P-well region PW corresponding to the region B. Consequently, the n-type ion implantation region **165**. In addition, arsenic ions are implanted into the polysilicon gate layer **115**, so that the n-type polysilicon gate layer is formed.

[0048] After the high-energy arsenic ions are impinged on the polysilicon gate layer **115**, many defects are generated in the polysilicon gate layer **115**. As mentioned above, the polysilicon gate layer **115** of the gate structure **125** is the floating gate of the floating gate transistor MF for storing the injected electrons. Due to the defects in the polysilicon gate layer **115**, a leakage path is generated. Consequently, the stored electrons are escaped from the floating gate, and the data retention capability of the memory cell is deteriorated.

[0049] In order to improve the data retention capability of the memory cell, the manufacturing process of the first embodiment needs to be modified. For example, in a second embodiment, the gate structure and the spacer are covered by a photoresist layer when the ion implantation process is performed.

[0050] Consequently, the defects in the polysilicon gate layer will be reduced, and the data retention capability of the memory cell will be increased. Since the manufacturing process is modified, the structure of the memory cell in the second embodiment is distinguished from the structure of the memory cell in the first embodiment.

[0051] FIGS. 3A to 3C schematically illustrate the steps of a method of manufacturing a single-poly non-volatile memory cell according to a second embodiment of the present invention. The steps of this embodiment before the step shown in FIG. 1H are similar to those of the first embodiment. For succinctness, only the steps after the step shown in FIG. 1H will be described as follows.

[0052] Please refer to FIG. 3A. After the photoresist layer **134** shown in FIG. 1H is removed, a photoresist layer **336** indicated by dashed lines is formed to cover the region A, the overlying gate structures **123**, **125** and the overlying spacers **153**, **155**. The photoresist layer **336** also covers the gate structure **125** and the spacer **155** corresponding to the region B. In addition, a smaller portion of the surface of the region B beside the spacer **155** is covered by the photoresist layer **336**, and a larger portion of the surface of the region B is exposed.

[0053] Then, an n-type ion implantation process is performed on the region B. Consequently, an n-type ion implantation region **365** indicated by oblique lines is formed in the region B uncovered by the photoresist layer **336**. Especially, the dopant concentration of the n-type ion implantation region **365** is greater than the dopant concentration of the NLDD region **145**.

[0054] FIG. 3B is a schematic cross-sectional view illustrating an enlarged portion of the structure shown in FIG. 3A and near the region B. As shown in FIG. 3B, a first portion of the NLDD region **145** and a portion of the n-type ion implantation region **365** are overlapped with each other, and a second portion of the NLDD region **145** including the portions directly and not directly below the spacer **155** is not overlapped with the n-type ion implantation region **365**. In addition, the NLDD region **145** and the n-type ion implantation region **365** are collaboratively formed as a merged n-doped region **375**. In addition, the merged n-doped region **375** is located beside the gate structure **125**.

[0055] When the n-type ion implantation process is performed, a small portion of the surface of the region B beside the spacer **155** is covered by the photoresist layer **336**. Consequently, in the x-direction, the width of the second portion of the NLDD region **145** in the merged n-doped region **375** is larger than the width of the spacer **155**. As shown in FIG. 3B, the second portion of the NLDD region **145** in the merged n-doped region **375** has a width $L_{sub.1}$, and the spacer **155** has a width $L_{sub.2}$, wherein $L_{sub.1}$ is greater than $L_{sub.2}$. For example, the width $L_{sub.1}$ is greater than $0.5\ \mu\text{m}$ and less than $5\ \mu\text{m}$, and the width $L_{sub.2}$ is greater than $0\ \mu\text{m}$ and less than $0.5\ \mu\text{m}$. It should be noted that embodiments are not limited to the width of the photoresist layers **336** over the gate structure **125** shown in FIG. 3A, as other photoresist layers may be used in some embodiments. For example, the width of the photoresist layers **336** over the gate structure **125** can be equal to the width of the polysilicon gate layer **115** plus the width of the spacers such that the width $L_{sub.1}$ is equal to the width $L_{sub.2}$.

[0056] Furthermore, when the n-type ion implantation process is performed, the gate structure **125** corresponding to the region B is covered by the photoresist layer **336**. Consequently, the dopant concentration of the polysilicon gate layer **115** is less than the dopant concentration of the merged n-doped region **375**. When compared with the first embodiment, the gate structure **125** corresponding to the region B is covered by the photoresist layer **336** according to the second embodiment. After the n-type ion implantation process is completed, the defects in the polysilicon gate layer **115** will be largely reduced, and the data retention capability of the memory cell can be

effectively enhanced.

[0057] Please refer to FIG. 3C. After the photoresist layer **336** is removed and the conducting line forming step is completed, the memory cell of the second embodiment is fabricated. That is, the merged p-type doped region **171** is connected with the source line SL, the merged p-type doped region **173** is connected with the bit line BL, the merged n-type doped region **375** is connected with the erase line EL, and the polysilicon gate layer **113** is connected with the select gate line SG.

[0058] The equivalent circuit of the memory cell in the second embodiment is identical to the equivalent circuit of the memory cell in the first embodiment.

[0059] The bias voltages of performing the program action, the erase action and the read action are similar, and not redundantly described herein.

[0060] FIGS. 4A to 4D schematically illustrate the steps of a method of manufacturing a single-poly non-volatile memory cell according to a third embodiment of the present invention. The steps of this embodiment before the step shown in FIG. 1H are similar to those of the first embodiment. For succinctness, only the steps after the step shown in FIG. 1H will be described as follows.

[0061] Please refer to FIG. 4A. After the photoresist layer **134** shown in FIG. 1H is removed, a photoresist layer **436** indicated by dashed lines is formed to cover the region A, the overlying gate structures **123**, **125** and the overlying spacers **153**, **155**. The photoresist layer **436** also covers the gate structure **125** and the spacer **155** corresponding to the region B. In addition, a smaller portion of the surface of the region B beside the spacer **155** is covered by the photoresist layer **436**, and a larger portion of the surface of the region B is exposed. It should be noted that embodiments are not limited to the width of the photoresist layers **336**, **436** over the gate structure **125** shown in FIG. 3A and FIG. 4A, as other photoresist layers may be used in some embodiments. For example, the width of the photoresist layers **336**, **436** over the gate structure **125** can be equal to the width of the polysilicon gate layer **115**.

[0062] Then, a first n-type ion implantation process is performed on the region B. Consequently, an n-type ion implantation region **465** indicated by oblique lines is formed in the region B uncovered by the photoresist layer **436**. Especially, the dopant concentration of the n-type ion implantation region **465** is greater than the dopant concentration of the NLDD region **145**.

[0063] Please refer to FIG. 4B. After the photoresist layer **436** is removed,

[0064] another photoresist layer **438** is formed. The photoresist layer **438** covers the region A, the overlying gate structures **123**, **125** and the overlying spacers **153**, **155**.

[0065] Then, a second n-type ion implantation process is performed on the region B by using the gate structure **125** and the spacer **155** as hard masks. Consequently, a second n-type ion implantation region **467** indicated by oblique lines are formed in the region B uncovered by the gate structure **125** and the spacer **155**.

[0066] In this embodiment, the provided energy in the first n-type ion implantation process is greater than the provided energy in the second n-type ion implantation process. Consequently, the depth of the first n-type ion implantation region **465** is greater than the depth of the second n-type ion implantation region **467**.

[0067] FIG. 4C is a schematic cross-sectional view illustrating an enlarged portion of the structure shown in FIG. 4B and near the region B. As shown in FIG. 4C, the NLDD region **145**, the first n-type ion implantation region **465** and the second n-type ion implantation region **467** are collaboratively formed as a merged n-doped region **475**. The merged n-doped region **475** is located beside the gate structure **125**. In addition, a first portion of the NLDD region **145** in the merged n-doped region **475** is overlapped with one of the n-type ion implantation region **465**, **467**, and a second portion of the NLDD region **145** in the merged n-doped region **475** is not overlapped with the first n-type ion implantation region **465**, and the second portion of the NLDD region **145** in the merged n-doped region **475** is not overlapped with the second n-type ion implantation region **467**.

[0068] As mentioned above, the gate structure **125** and the spacer **155** are used as the masks when the second n-type ion implantation process is performed. Consequently, the width of the second

portion of the NLDD region **145** in the merged n-doped region **475** is equal to the width of the spacer **155**.

[0069] As mentioned above, the gate structure **125** corresponding to the region B is covered by the photoresist layer **436** when the first n-type ion implantation process is performed. Consequently, after the second n-type ion implantation process is completed, it is assured that the dopant concentration of the polysilicon gate layer **115** is substantially equal to the dopant concentration of the second n-type ion implantation region **467** and the dopant concentration of the polysilicon gate layer **115** is less than the dopant concentration of the merged n-doped region **475**. For example, the dopant concentration of the polysilicon gate layer **115** is A (particle/cm.sup.3), and the dopant concentration of the merged n-doped region **475** is B (particle/cm.sup.3). Preferably, $A < (B \times 40\%)$. That is, the dopant concentration of the polysilicon gate layer **115** is less than 40% of the dopant concentration of the merged n-doped region **475**.

[0070] Furthermore, when the second n-type ion implantation process is performed, less energy is provided. Consequently, the defects in the polysilicon gate layer **115** will be largely reduced when compared with the first embodiment and the third embodiment. Consequently, the data retention capability of the memory cell can be effectively enhanced.

[0071] Please refer to FIG. **4D**. After the photoresist layer **438** is removed and the conducting line forming step is completed, the memory cell of the third embodiment is fabricated. That is, the merged p-type doped region **171** is connected with the source line SL, the merged p-type doped region **173** is connected with the bit line BL, the merged n-type doped region **475** is connected with the erase line EL, and the polysilicon gate layer **113** is connected with the select gate line SG.

[0072] The equivalent circuit of the memory cell in the third embodiment is identical to the equivalent circuit of the memory cell in the first embodiment. The bias voltages of performing the program action, the erase action and the read action are similar, and not redundantly described herein.

[0073] From the above descriptions, the present invention provides the single-poly non-volatile memory cell. In the memory cell, the polysilicon gate layer **115** of the gate structure **125** is used as the floating gate of the floating gate transistor. Furthermore, during the ion implantation process, a relatively small amount of dopant is implanted into the polysilicon gate layer **115**. Consequently, the dopant concentration of the polysilicon gate layer **115** is less than the dopant concentration of the merged n-type doped region **475** beside the gate structure **125**. Since the defects in the polysilicon gate layer **115** are largely reduced, the data retention capability of the memory cell is effectively enhanced.

[0074] In the above embodiments, the select transistor M.sub.S and the floating gate transistor M.sub.F are p-type transistors. It is noted that the types of these transistors are not restricted. For example, in another embodiment, the manufacturing process of the memory cell is modified, and thus the select transistor M.sub.S and the floating gate transistor M.sub.F are n-type transistors.

[0075] For example, in the structure of FIG. **1K**, the first well region under the surface of the region A is modified as a P-well region PW, and the merged doped regions **171**, **172** and **173** are modified as merged n-type doped regions. Consequently, the select transistor M.sub.S and the floating gate transistor M.sub.F are n-type transistors. Similarly, in case that the floating gate transistor M.sub.F is the n-type transistor, the technologies of the second embodiment and the third embodiment can be applied in the process of forming the merged doped regions **171**, **172** and **173** corresponding to the region A. Since the defects in the polysilicon gate layer **115** of the gate structure **125** is largely reduced, the data retention capability of the memory cell is effectively enhanced.

[0076] While the invention has been described in terms of what is presently considered to be the most practical and preferred embodiments, it is to be understood that the invention needs not be limited to the disclosed embodiment. On the contrary, it is intended to cover various modifications and similar arrangements included within the spirit and scope of the appended claims which are to

be accorded with the broadest interpretation so as to encompass all such modifications and similar structures.

Claims

1. A single-poly non-volatile memory cell, comprising: a substrate; an isolation structure formed on the substrate; a first well region formed on the substrate; a second well region formed on the substrate, wherein the isolation structure is located between the first well region and the second well region; a first gate structure and a second gate structure formed on the first well region, wherein the second gate structure is extended to the second well region through a surface of the isolation structure, and a portion of the second well region is covered by the second gate structure; a first spacer formed on a sidewall of the first gate structure; a second spacer formed on a sidewall of the second gate structure; a first merged doped region formed under a surface of the first well region, and located beside a first side of the first gate structure; a second merged doped region formed under the surface of the first well region, and arranged between a second side of the first gate structure and a first side of the second gate structure; a third merged doped region formed under the surface of the first well region, and located beside a second side of the second gate structure; and a fourth merged doped region formed under a surface of the second well region, and located beside the second gate structure, wherein the first gate structure comprises a first gate oxide layer and a first polysilicon gate layer, the second gate structure comprises a second gate oxide layer and a second polysilicon gate layer, and a dopant concentration of the second polysilicon gate layer above the second well region is less than a dopant concentration of the fourth merged doped region.
2. The single-poly non-volatile memory cell as claimed in claim 1, wherein the first merged doped region, the first gate structure and the second merged doped region are collaboratively formed as a select transistor, the second merged doped region, the second gate structure and the third merged doped region are collaboratively formed as a floating gate transistor, and the second gate structure and the fourth merged doped region are collaboratively formed as a MOS capacitor.
3. The single-poly non-volatile memory cell as claimed in claim 2, wherein a gate terminal of the select transistor is connected with a select gate line, a first drain/source terminal of the select transistor is connected with a source line, a first drain/source terminal of the floating gate transistor is connected with a second drain/source terminal of the select transistor, a second drain/source terminal of the floating gate transistor is connected with a bit line, a first terminal of the MOS capacitor is connected with a floating gate of the floating gate transistor, and a second terminal of the MOS capacitor is connected with an erase line.
4. The single-poly non-volatile memory cell as claimed in claim 2, wherein the select transistor and the floating gate transistor are p-type transistors, the second gate structure and the fourth merged doped region are collaboratively formed as an n-type transistor, and a first drain/source terminal and a second drain/source terminal of the n-type transistor are connected with each other to form an n-type MOS capacitor.
5. The single-poly non-volatile memory cell as claimed in claim 2, wherein the second gate structure and the fourth merged doped region are collaboratively formed as a first transistor, and the first transistor, the select transistor and the floating gate transistor are n-type transistors, wherein a first drain/source terminal and a second drain/source terminal of the first transistor are connected with each other to form an n-type MOS capacitor.
6. The single-poly non-volatile memory cell as claimed in claim 1, wherein the fourth merged doped region comprises a lightly doped drain region and an ion implantation region, wherein a first portion of the lightly doped drain region is overlapped with a portion of the ion implantation region, and a second portion of the lightly doped drain region is not overlapped with the ion implantation region.

7. The single-poly non-volatile memory cell as claimed in claim 6, wherein the fourth merged doped region is a merged n-type doped region, the lightly doped drain region is an n-type lightly doped drain region, and the ion implantation region is an n-type ion implantation region.
8. The single-poly non-volatile memory cell as claimed in claim 6, wherein a width of the second portion of the lightly doped drain region is equal to or greater than a width of the second spacer.
9. The single-poly non-volatile memory cell as claimed in claim 8, wherein the width of the second portion of the lightly doped drain region is greater than 0.5 μm and less than 5 μm , and the width of the second spacer is greater than 0 μm and less than 0.5 μm .
10. The single-poly non-volatile memory cell as claimed in claim 1, wherein the fourth merged doped region comprises a lightly doped drain region, a first ion implantation region and a second ion implantation region, wherein a first portion of the lightly doped drain region is overlapped with one of the first ion implantation region and the second ion implantation region, and a second portion of the lightly doped drain region is not overlapped with the first ion implantation region, and the second portion of the lightly doped drain region is not overlapped with the second ion implantation region.
11. The single-poly non-volatile memory cell as claimed in claim 10, wherein the fourth merged doped region is a merged n-type doped region, the first lightly doped drain region is a first n-type lightly doped drain region, the first ion implantation region is a first n-type ion implantation region, and the second ion implantation region is a second n-type ion implantation region.
12. The single-poly non-volatile memory cell as claimed in claim 10, wherein a width of the second portion of the lightly doped drain region is equal to a width of the second spacer.
13. The single-poly non-volatile memory cell as claimed in claim 10, wherein a depth of the first ion implantation region is greater than a depth of the second ion implantation region, a dopant concentration of the second polysilicon gate layer above the second well region is equal to a dopant concentration of the second ion implantation region, and the dopant concentration of the second polysilicon gate layer above the second well region is less than a dopant concentration of the merged n-doped region.
14. The single-poly non-volatile memory cell as claimed in claim 13, wherein the dopant concentration of the second polysilicon gate layer above the second well region is less than 40% of the dopant concentration of the fourth merged doped region.
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